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EE1030/EE2100: Matrix Theory

Lecture 33: Eigenvalues and Eigenvectors

Topics :

1. Properties of Eigenvalues and Eigenvectors
2. Generalized Eigenvalues

Properties of Eigenvalues and Eigenvectors

$\langle a, \vec{b} \rangle = \sum_{i=1}^{n} a_i b_i^*$

$\langle \vec{b}, \vec{a} \rangle = (\quad)^*$

Prop: If $A \in \mathbb{R}^{n \times n}$ is symmetric, then all eigenvalues are real.

Proof:- λ is the eig val with associated eig vec. $\vec{v}$ (allowing $\lambda \in \mathbb{C}$ & $\vec{v} \in \mathbb{C}$)

$$A\vec{v} = \lambda \vec{v} \longrightarrow (1)$$

Premult (1) with $(\vec{v}^*)^T \Rightarrow (\vec{v}^*)^T A \vec{v} = \lambda \left[(\vec{v}^*)^T \vec{v} \right] \longrightarrow (2)$ ($\in \mathbb{R}$)

$(\vec{v}^*)^T A \vec{v}$ ↓ Scalar.

Take transpose of $(\vec{v}^*)^T A \vec{v} = \vec{v}^T A^T \vec{v}^* = \vec{v}^T A \vec{v}^* \longrightarrow (3)$

Since $(\vec{v}^*)^T A \vec{v}$ is a scalar, its transpose must be same as its value

$$(\vec{v}^*)^T A \vec{v} = \vec{v}^T A (\vec{v}^*)$$

$$= \left((\vec{v}^*)^T A \vec{v} \right)^* \Rightarrow (\vec{v}^*)^T A \vec{v} \in \mathbb{R}. \longrightarrow (4)$$

from (4) & (2) $\Rightarrow \lambda \in \mathbb{R}$.

$\Rightarrow$ does not rule out the possibility that $\vec{v} \in \mathbb{C}^n \Rightarrow A\vec{v} = \lambda \vec{v}$

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If $\vec{v} = \vec{w} + j\vec{u} \Rightarrow A(\vec{w} + j\vec{u}) = \lambda(\vec{w} + j\vec{u})$

$\Rightarrow \vec{w}$ & $\vec{u}$ are eig vec of A.

Properties of Eigenvalues and Eigenvectors

Prop: If $A \in \mathbb{R}^{n\times n}$ is symmetric, then eig vec corresponding to distinct eig val are orthogonal.

Proof:- Let λ_1, λ_2 denote distinct eig val. with associated eig vec $\vec{v}_1$ & $\vec{v}_2$

$$A\vec{v}_1 = \lambda_1\vec{v}_1 \rightarrow (1) \quad \& \quad A\vec{v}_2 = \lambda_2\vec{v}_2 \rightarrow (2)$$

$\lambda \in \mathbb{R}$, $\because A$ is symmetric

$\vec{v}_1, \vec{v}_2$ can be chosen to be real.

Pre mult (1) with $\vec{v}_2^T$ and (2) with $\vec{v}_1^T$

$$\vec{v}_2^T A\vec{v}_1 = \lambda_1 \vec{v}_2^T\vec{v}_1 \rightarrow (3)$$

$$\& \quad \vec{v}_1^T A\vec{v}_2 = \lambda_2 \vec{v}_1^T\vec{v}_2 \rightarrow (4)$$

$\because \vec{v}_2^T A\vec{v}_1 \; \& \; \vec{v}_1^T A\vec{v}_2 \in \mathbb{R}$, $\Rightarrow$ Transpose of these QF must be the same as the orig. val.

$$(\vec{v}_2^T A\vec{v}_1)^T = \vec{v}_2^T A\vec{v}_1$$

$$\vec{v}_1^T A^T\vec{v}_2 = \vec{v}_2^T A\vec{v}_1 \rightarrow (5)$$

Sub (5) in (3) $\quad \vec{v}_1^T A\vec{v}_2 = \lambda_1 \vec{v}_2^T\vec{v}_1 \rightarrow (6)$ &

$$\vec{v}_1^T A\vec{v}_2 = \lambda_2 \vec{v}_1^T\vec{v}_2 \rightarrow (4)$$

$$\Rightarrow \quad (\lambda_1 - \lambda_2)\,\vec{v}_1^T\vec{v}_2 = 0 \Rightarrow \vec{v}_1^T\vec{v}_2 = 0$$

$\because \lambda_1 \neq \lambda_2$ & $\vec{v}_i \neq \vec{0}$

Algebraic Multiplicity and Geometric Multiplicity

given $A \in \mathbb{R}^{n \times n}$, the characteristic polynomial of A : $P_A(\lambda) = \det(A - \lambda I) = 0$

(m_a) alg mul of λ_i = no. of times it repeats (roots of the characteristic polynomial)

for every eig val E_λ = Span { linearly independent eig vec }

↑ eigen space

$\dim(E_\lambda)$ = geo. mul. of λ. (m_g)

Ex: $A = \begin{bmatrix} 2 & 1 \\ 0 & 2 \end{bmatrix}$

eig values = 2, 2, alg. mul of 2 = 2.

E_2 ← eigen space. = Span $\{\vec{v}_1\}$ where $\vec{v}_1 = \begin{bmatrix} 1 \\ 0 \end{bmatrix}$ ⇒ geo mul of $\lambda = 2$ is 1

geo mul < alg. mul

$A = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$

eig val = 1, 1 alg mul of 1 = 2.

$E_1 = \mathbb{R}^2$ but one choice of basis for $E_1 = (\vec{e}_1 \& \vec{e}_2)$

geo. mul of $\lambda = 1$ is 2.

Algebraic Multiplicity and Geometric Multiplicity

Prop: for every eig val of $A \in \mathbb{R}^{n\times n}$, geo. mul $\leq$ alg. mul ($m_g \leq m_a$)

Proof:- Say geo mul of λ_i is $k = m_g \Rightarrow \exists (\vec{v}_1, \dots \vec{v}_k)$ that are LI & associated with λ_i

form $P = [\vec{v}_1, \dots \vec{v}_k, \underbrace{\vec{v}_{k+1}, \dots \vec{v}_n}_{\hookrightarrow \text{are other LI vectors from } \mathbb{R}^n}] \Rightarrow \text{Rank}(P) = n \Rightarrow \det(P) \neq 0$

$$(A-\lambda I)P = AP - \lambda P$$

$$= A[\vec{v}_1, \dots \vec{v}_k, \vec{v}_{k+1}, \dots \vec{v}_n] - \lambda P$$

$$= [\lambda_i \vec{v}_1 \dots \lambda_i \vec{v}_k, A\vec{v}_{k+1}, \dots A\vec{v}_n] - \lambda [\vec{v}_1, \dots \vec{v}_n]$$

$$= [(\lambda_i-\lambda)\vec{v}_1, \dots, (\lambda_i-\lambda)\vec{v}_k, \underbrace{A\vec{v}_{k+1} - \lambda\vec{v}_{k+1}}_{\vec{\omega}_{k+1}(\lambda)}, \dots \underbrace{A\vec{v}_n - \lambda\vec{v}_n}_{\vec{\omega}_n(\lambda)}]$$

$$\det[(A-\lambda I)P] = (\lambda_i-\lambda)^k \underbrace{\det[\vec{v}_1, \dots \vec{v}_k, \vec{\omega}_{k+1}(\lambda) \dots \vec{\omega}_n(\lambda)]}_{\phi(\lambda)}$$

$$\det(A-\lambda I)\det(P) = (\lambda_i-\lambda)^k \phi(\lambda) \quad (\det(P) \neq 0)$$

$$\Rightarrow \det(A-\lambda I) = (\lambda_i-\lambda)^k \frac{\phi(\lambda)}{\det(P)} \Rightarrow \text{eig val of } A \text{ (determined by } \det(A-\lambda I)=0)$$

Generalized Eigenvectors

$\det(A-\lambda I) = (\lambda_i - \lambda)^k \frac{\phi(\lambda)}{\det(\phi)} \Rightarrow$ eig val of A (determined by $\det(A-\lambda I)=0$)

will have λ_i atleast k times.

will be $> k$ if $\phi(\lambda_i) = 0$

$\Rightarrow$ alg mul of λ_i is atleast k.

$\Rightarrow$ alg mul $\geq$ geo. mul.